

Lecture 6 — Network RM: Fluid LPs and Bid Prices

*Lecturer: Sid Banerjee**Date: September 10, 2026*

6.1 Overview

Lecture 5 gave an exact acceptance rule for a request:

accept current reward $\iff$ reward exceeds displaced future value.

With one resource, dynamic programming can compute the opportunity cost of capacity exactly. In a network, the state is a vector of remaining capacities and the exact dynamic program becomes exponentially large.

This transition illustrates a recurring tension:

Scale enlarges the state space and defeats exact state-by-state optimization,
while aggregate regularity makes a fluid approximation meaningful.

The central question is:

How can we approximate the opportunity cost of many shared resources?

Here is the progression from Lecture 5's exact marginal value to a scalable control:

offline packing $\longrightarrow$ fluid LP $\longrightarrow$ dual resource prices $\longrightarrow$ bid-price control.

Sections marked * are optional reading.

6.2 The network revenue-management model

There are m resources with remaining capacity

$$b = (b_1, \dots, b_m) \in \mathbb{Z}_+^m.$$

There are n request types. A type- j request:

- earns reward r_j if accepted;

- consumes resource vector $a_j \in \{0, 1, \dots\}^m$;
- earns and consumes nothing if rejected.

Collect the consumption vectors into the matrix

$$A = [a_1 \ a_2 \ \cdots \ a_n].$$

Examples include an airline itinerary using several flight legs, a hotel stay using a room on several nights, an ad impression consuming several campaign budgets, or an order drawing inventory from multiple warehouses.

If type j arrives with probability λ_j in each period, then with τ periods remaining its arrival count is

$$Z_j(\tau) = \sum_{s=1}^{\tau} \mathbf{1}\{J_s = j\}, \quad \mathbb{E}[Z_j(\tau)] = \tau \lambda_j.$$

6.3 A running two-leg airline example

Keep the following network in mind. There are two flight legs with one remaining seat on each. A local itinerary on either leg earns 6 and consumes one seat; a connecting itinerary earns 9 and consumes one seat on each leg. The connecting request has the largest individual reward, but accepting it may displace two local requests worth 12 together. This is the essential difficulty of network scarcity: a request must be evaluated against the combined opportunities it consumes.

6.4 What would a clairvoyant do?

Suppose the entire future count vector is revealed to an *offline clairvoyant*:

$$Z = (Z_1, \dots, Z_n).$$

The offline clairvoyant chooses how many requests of each type to accept:

$$\begin{aligned} \text{OPT}(Z, b) = \max_{x \in \mathbb{Z}_+^n} \quad & \sum_{j=1}^n r_j x_j \\ \text{s.t.} \quad & Ax \leq b, \\ & 0 \leq x_j \leq Z_j \quad \forall j. \end{aligned} \tag{6.1}$$

This offline optimum has strictly more information than any online policy, so it is a sample-path upper bound on achievable revenue. It also exposes the allocation problem hidden inside revenue management: which combination of requests makes the best use of shared capacity?

For many RM problems, we first relax integrality. The resulting LP remains clairvoyant because it still depends on the unknown vector Z .

6.5 The fluid LP

Replace future type counts by their expectations

$$d_j = \mathbb{E}[Z_j(\tau)] = \tau \lambda_j$$

and relax integrality. The *fluid* or deterministic LP is

$$\begin{aligned} \text{FLUID}(d, b) = \max_{x \geq 0} \quad & r^\top x \\ \text{s.t.} \quad & Ax \leq b, \\ & x \leq d. \end{aligned} \tag{6.2}$$

Its interpretation is aggregate rather than sequential:

If future demand arrived exactly at its mean, how many requests of each type would we serve?

The fluid LP removes two difficulties at once: it replaces a random future by deterministic expected counts, and it replaces a dynamic control problem by a static allocation problem.

It is important to keep four objects separate:

offline clairvoyant problem:	uses realized future counts Z ;
fluid LP:	uses expected counts $d = \mathbb{E}[Z]$;
fluid primal solution:	predicts aggregate acceptance counts;
fluid dual solution:	prices scarce resource capacity.

6.5.1 Further perspective: why the fluid value is an upper bound

Let $v(z, b)$ denote the optimal value of the *relaxed* offline LP with demand upper bound z . The LP value is concave in its right-hand side z : mixing two feasible demand environments cannot produce less value than mixing their optimal allocations. Jensen's inequality therefore gives

$$\mathbb{E}[v(Z, b)] \leq v(\mathbb{E}[Z], b) = \text{FLUID}(\mathbb{E}[Z], b).$$

After accounting for the integrality relaxation,

$$\mathbb{E}[\text{OPT}(Z, b)] \leq \text{FLUID}(\mathbb{E}[Z], b).$$

Remark 6.1. This is not a generic rule that “optimization at an expectation” upper-bounds “the expectation of optimization.” The direction comes from the concavity of this particular relaxed LP value as a function of the demand right-hand side.

6.6 Solving the two-leg example

Write the three products from our running example as:

product	a_j	r_j
1 : leg 1 only	(1, 0)	6
2 : leg 2 only	(0, 1)	6
3 : connecting itinerary	(1, 1)	9.

Suppose the expected future count of each product is one. The fluid LP is

$$\begin{aligned} \max \quad & 6x_1 + 6x_2 + 9x_3 \\ \text{s.t.} \quad & x_1 + x_3 \leq 1, \\ & x_2 + x_3 \leq 1, \\ & 0 \leq x_j \leq 1. \end{aligned}$$

The optimal solution is

$$x^* = (1, 1, 0), \quad \text{FLUID} = 12.$$

Although the connecting itinerary has the largest individual reward, it uses both seats and displaces two local products worth 12 together.

This is network scarcity: the cost of a request is the combined opportunity cost of every resource it consumes.

6.7 Duality and bid prices

Associate resource-price vector $p \geq 0$ with $Ax \leq b$ and demand-price vector $z \geq 0$ with $x \leq d$. The dual of (6.2) is

$$\min_{p, z \geq 0} \quad b^\top p + d^\top z \tag{6.3}$$

$$\text{s.t.} \quad a_j^\top p + z_j \geq r_j \quad \forall j. \tag{6.4}$$

The resource dual p_i is the marginal fluid value of one additional unit of resource i . In revenue management it is called a *bid price*.

For the two-leg example, one optimal dual solution is

$$p = (5, 5), \quad z = (1, 1, 0).$$

The dual objective is $10 + 2 = 12$, matching the primal value. The resource cost of each product is

product	$p^\top a_j$	r_j
1	5	6
2	5	6
3	10	9.

Before writing a rule, read the table economically. A local request pays 6 for a resource whose approximate opportunity cost is 5, so it creates positive net value. The connecting request pays 9 but consumes resources worth about 10 together, so accepting it would destroy more future opportunity than it creates.

This suggests the bid-price rule

$$\boxed{\text{accept type } j \text{ if capacity is feasible and } r_j \geq p^\top a_j.} \quad (6.5)$$

It accepts each local itinerary and rejects the connecting itinerary, exactly as the primal allocation suggests.

Remark 6.2 (A callback to Lecture 1). *In Lecture 1, dual prices coordinated buyers competing for heterogeneous goods. Here, dual prices coordinate requests competing for shared resources. In both cases, a centralized allocation problem produces prices that support decentralized decisions.*

6.8 Exact and approximate opportunity costs

We have now seen the value of capacity in two forms. The exact dynamic opportunity cost of accepting request j in state (t, b) is

$$V_{t-1}(b) - V_{t-1}(b - a_j).$$

It is economically correct but usually computationally intractable.

The fluid approximation is

$$p^\top a_j.$$

It decomposes across resources and is easy to use online, but it comes from a deterministic approximation.

The primal solution x^* carries complementary information: it predicts how many requests of each type an offline optimum should accept in aggregate. Thus the fluid LP provides both

dual prices p : approximate opportunity costs, primal counts x^* : predicted offline action counts.
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Lecture 7 will focus on the second interpretation.

6.9 From an LP to an online policy

The classical bid-price implementation solves the fluid LP, extracts p , and applies (6.5) to each feasible arrival. One static optimization has become a sequential control rule.

6.9.1 Optional: static randomized control *

If the fluid plan accepts x_j^* requests from an expected d_j arrivals, another implementation accepts each feasible type- j request with probability

$$q_j = \frac{x_j^*}{d_j}.$$

This reproduces the fluid mix in expectation but treats x^* as an exact point forecast.

6.9.2 Optional: re-solving *

With τ periods and capacity b remaining, one can periodically solve

$$\text{FLUID}(\tau\lambda, b)$$

again. Re-solving adjusts to realized scarcity and updates bid prices. It still raises a statistical question: how uncertain is the new fluid prediction?

6.10 What is missing?

The fluid LP predicts future counts using their means. But typical deviations have scale

$$Z_j(\tau) - \tau\lambda_j = O_{\mathbb{P}}(\sqrt{\tau}).$$

This does not make the LP a bad benchmark. It means that its output should be treated as a prediction with uncertainty, not as an exact forecast.

Lecture 7 asks:

How much of the fluid prediction is safe enough to act on?

6.11 Takeaways

1. Network RM is an online packing problem with shared resource constraints.
2. The clairvoyant offline allocation is a natural sample-path benchmark.
3. The fluid LP replaces random future counts by their expectations.
4. Its dual resource variables are bid prices: approximate marginal values of capacity.
5. Its primal solution predicts aggregate offline action counts.
6. The next issue is statistical: a fluid prediction comes with uncertainty of order $\sqrt{\tau}$.

Recommended reading

Core / closest to this lecture.

Talluri and van Ryzin, Sections 3.1.2.3 and 3.2.2–3.2.5. Read for bid-price controls, their limitations, and the interpretation of bid prices as opportunity costs.

Alternative perspective.

Talluri and van Ryzin, Section 3.3.1, develops the deterministic LP approximation. Compare its primal allocation targets with its dual resource prices; these are the two fluid outputs emphasized here.

Optional / deeper.

Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, reframes online allocation around prediction of a dynamic offline benchmark. It is research-level reading; at this stage, focus on why the clairvoyant allocation is a useful benchmark rather than on the proofs.